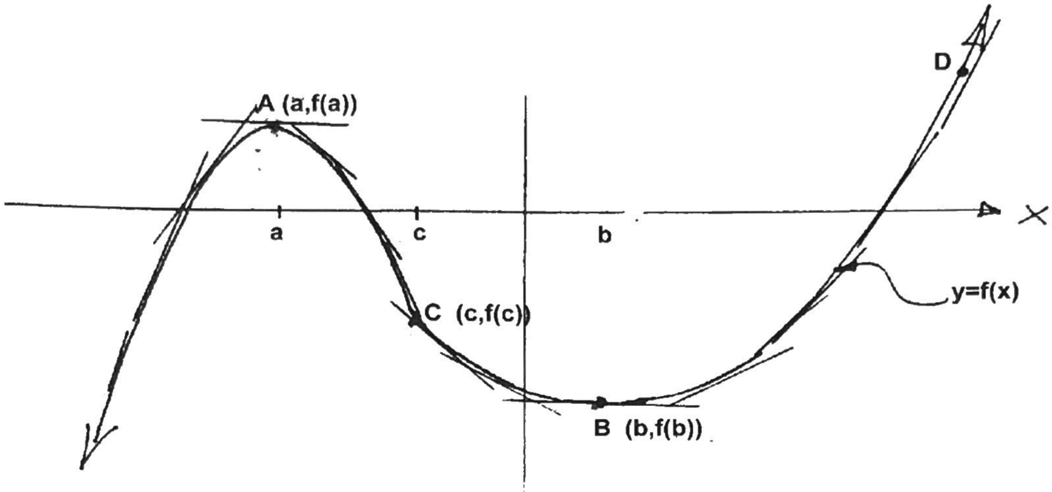
Maximum and Minimum Values

Knowledge of the slope of the tangent line to a curve (function) allows us to pinpoint locations of local maxima or minima. Also known as local extrema.



Point A (on the graph above) is termed a **relative or local maximum** point because in A's vicinity there are no points higher. D is higher but it is not in the vicinity of A. Similarly B is known as a **relative or local minimum** point, because in B's vicinity there is no point lower.

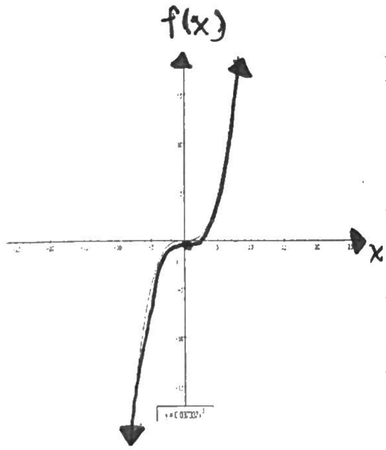
Critical Number: A number c in the domain of f is a critical number of f if either $f'(c)=0$ or $f'(c)$ does not exist. The point $(c, f(c))$ is called a **critical point**.

Fermat's Theorem: If f has a local maximum or minimum value at $x = c$, then either $f'(c)=0$ **or** $f'(c)$ does not exist.

<u>1st Derivative Test</u>

For a local maximum to occur, $f'(x)$ must change from positive to negative.

For a local minimum to occur, $f'(x)$ must change from negative to positive.



Note: Note the sign of $f'(x)$ must change for a local extrema.

Example: Consider $f(x) = x^3$. $(f'(x) = 3x^2)$

When $x = 0$, $f'(x) =$ 0 .

Yet (0,0) is obviously not a relative maximum nor minimum point. It may at best be described as a *"flat spot"*. Mathematically, it is called a "point of inflection" (POI).

Maximum and Minimum Values

Example 1: Determine the coordinates of any local extrema for the function $f(x)=x^4-x^3$.

$f'(x) = 4x^3 - 3x^2$

$= x^2(4x-3) \Rightarrow$ critical #'s @ $x = 0, \frac{3}{4}$

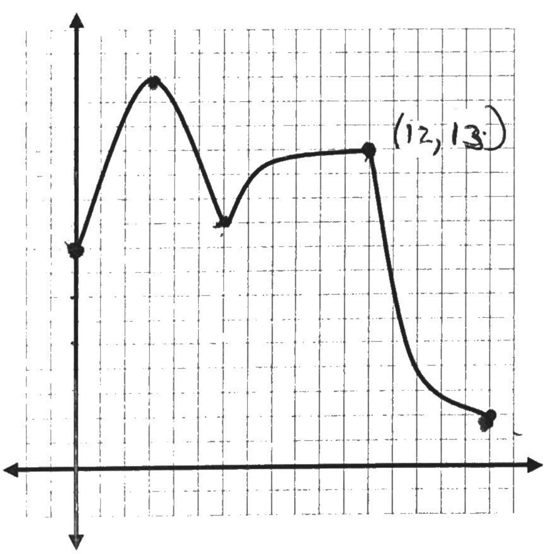
Interval	$(-\infty,0)$	$(0,\frac{3}{4})$	$(\frac{3}{4},\infty)$
x^2	+	+	+
$4x-3$	−	−	+
$f'(x)$	−	−	+
$f(x)$	↓	↓	↑

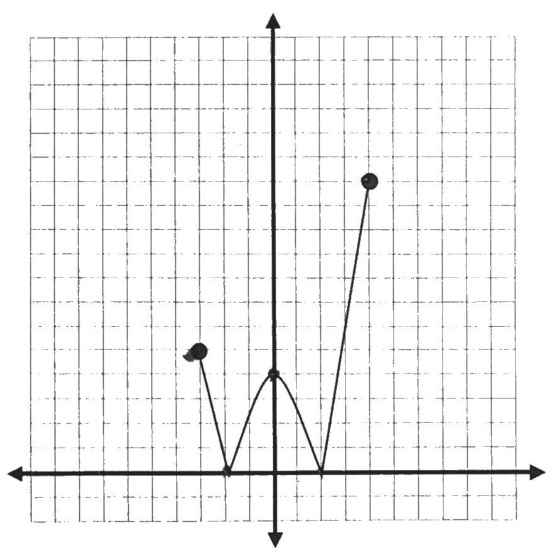
* $x=0$ is not a local extrema

* local min @ $(\frac{3}{4}, \frac{-27}{256})$

To find absolute maximum or minimum points, you need to also test the endpoints of the domain.

Example 2: Determine the local and absolute extrema for each function:





$y=f(x)$ for $x\in[0,12]$

Local Max: (3,16)

Local min: (6,10)

Abs. max: (3,16)

Abs. min: (0,9)

$y=|x^2-4|$ for $x\in[-3,4]$

Local max: (0,4)

Local min: (−2,0) & (2,0)

Abs. max: (4,12)

Abs. min: (−2,0) & (2,0)

Example 3: Determine the absolute maximum and absolute minimum values of the function $f(x) = 2x^4 - 4x^2 + 4$ for $x \in [-2, 0.5]$.

$f'(x) = 8x^3 - 8x$

$= 8x(x^2 - 1)$

1 is not a critical #.

$\Rightarrow$ critical #'s @ $x = 0, \pm 1$.

Interval	$[-2, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
x	$-$	$-$	$+$	$+$
$x^2 - 1$	$+$	$-$	$-$	$+$
$f'(x)$	$-$	$+$	$-$	$+$
$f(x)$	↓	↑	↓	↑

not in domain

local min @ $x = -1$

local max @ $x = 0$

Test

* $f(-1) = 2 \rightarrow$ Absolute min. @ $(-1, 2)$
* $f(0) = 4 \rightarrow$ Local Max.
* $f(1) = 2 \rightarrow$ NOT IN DOMAIN
* $f(-2) = 20 \rightarrow$ Absolute max. @ $(-2, 20)$
* $f(0.5) = \frac{25}{8}$

Maximum and Minimum Values

Additional Questions

1. Determine the local max./min. of $f(x) = 3x^5 - 5x^3$.

$f'(x) = 15x^4 - 15x^2$

$= 15x^2(x^2-1)$

$\therefore$ critical #'s: $x = 0, \pm 1$

Interval	$(-\infty,-1)$	$(-1,0)$	$(0,1)$	$(1,\infty)$
x^2	+	+	+	+
x^2-1	+	−	−	+
$f'(x)$	+	−	−	+
$f(x)$	↑	↓	↓	↑

local max. @ $(-1, 2)$

local min. @ $(1, -2)$

NOTE:

$(0,0)$ is not a local max/min. No change in $f'(x)$

2. Given $f(x) = ax^3 + bx^2 + cx + d$, determine the values of a, b, c and d such that there is a local maximum at $(-2, 13)$ and a local minimum at $(0, 5)$.

$f'(x) = 3ax^2 + 2bx + c$

sub. $x = -2$ into $f'(x)$ and set $= 0$.

$12a - 4b + c = 0$ ①

sub. $x = 0$ into $f'(x)$ and set $= 0$

$\boxed{c = 0}$

sub. $(0, 5)$ into $f(x)$

$\boxed{d = 5}$

sub. $(-2, 13)$ into $f(x)$

$-8a + 4b + 5 = 13$

$-8a + 4b = 8$ ②

① + ②

$4a = 8$

$\boxed{a = 2}$

sub. $a = 2$ into ②

$\boxed{b = 6}$